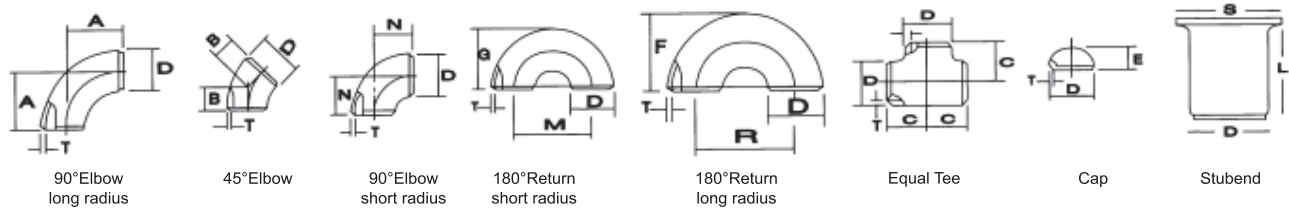


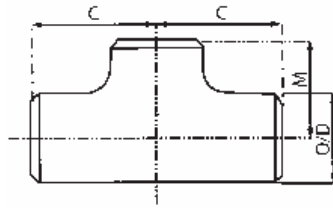
**DIMENSION OF BUTT WELD FITTINGS ANSI B-16.9 / B-16.29**



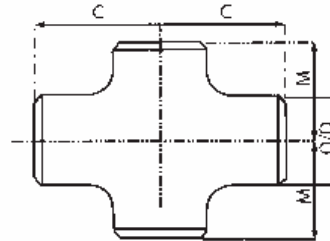
| Nominal Pipe Size |     | Outside Diameter | Center to Face |       |       |           | Back to Face |                                                                                      |       | Center to Center |        |       | Length 'L'         |                 |
|-------------------|-----|------------------|----------------|-------|-------|-----------|--------------|--------------------------------------------------------------------------------------|-------|------------------|--------|-------|--------------------|-----------------|
| INCH              | MM  | D                | A<br>R=1.5D    | B     | C     | N<br>R=1D | E            | F                                                                                    | G     | R                | M      | S     | Short L<br>MSSSP43 | Long L<br>B16.9 |
| 1/2               | 15  | 21.3             | 38.00          | 16.0  | 25.0  | -         | 25.0         | 48.0                                                                                 | -     | 76.0             |        | 35.0  | 50.8               | 76.2            |
| 3/4               | 20  | 26.7             | 29.00          | 11.0  | 29.0  | -         | 25.0         | 43.0                                                                                 | -     | 57.0             |        | 43.0  | 50.8               | 76.2            |
| 1                 | 25  | 33.4             | 38.00          | 22.0  | 38.0  | 25.0      | 38.0         | 56.0                                                                                 | 41.0  | 76.0             | 51.0   | 51.0  | 50.8               | 101.6           |
| 1.1/4             | 32  | 42.2             | 48.00          | 25.0  | 48.0  | 32.0      | 38.0         | 70.0                                                                                 | 52.0  | 95.0             | 64.0   | 64.0  | 50.8               | 101.6           |
| 1.1/2             | 40  | 48.3             | 57.15          | 29.0  | 57.0  | 38.0      | 38.0         | 83.0                                                                                 | 62.0  | 114.0            | 76.0   | 73.0  | 50.8               | 101.6           |
| 2                 | 50  | 60.3             | 76.00          | 35.0  | 64.0  | 51.0      | 38.0         | 106.0                                                                                | 81.0  | 152.0            | 102.0  | 93.0  | 63.5               | 152.4           |
| 2.1/2             | 65  | 73.0             | 95.25          | 44.0  | 76.0  | 64.0      | 38.0         | 132.0                                                                                | 100.0 | 191.0            | 127.0  | 105.0 | 63.5               | 152.4           |
| 3                 | 80  | 88.9             | 114.30         | 51.0  | 86.0  | 76.0      | 51.0         | 159.0                                                                                | 121.0 | 229.0            | 152.0  | 127.0 | 63.5               | 152.4           |
| 3.1/2             | 90  | 101.6            | 133.35         | 57.0  | 95.0  | 89.0      | 64.0         | 184.0                                                                                | 140.0 | 267.0            | 175.0  | 140.0 | 76.2               | 152.4           |
| 4                 | 100 | 114.3            | 152.0          | 64.0  | 105.0 | 102.0     | 64.0         | 210.0                                                                                | 159.0 | 305.0            | 203.0  | 157.0 | 76.2               | 152.4           |
| 5                 | 125 | 141.3            | 190.0          | 79.0  | 123.0 | 127.0     | 76.0         | 262.0                                                                                | 197.0 | 381.0            | 254.0  | 186.0 | 76.2               | 152.4           |
| 6                 | 150 | 168.3            | 229.0          | 95.0  | 143.0 | 152.0     | 89.0         | 313.0                                                                                | 237.0 | 457.0            | 305.0  | 218.0 | 88.9               | 203.2           |
| 8                 | 200 | 219.1            | 305.0          | 127.0 | 178.0 | 203.0     | 102.0        | 414.0                                                                                | 313.0 | 610.0            | 406.0  | 270.0 | 101.6              | 203.2           |
| 10                | 250 | 273.1            | 381.0          | 159.0 | 216.0 | 254.0     | 127.0        | 518.0                                                                                | 391.0 | 762.0            | 508.0  | 324.0 | 127.0              | 254.0           |
| 12                | 300 | 323.8            | 457.0          | 190.0 | 254.0 | 305.0     | 152.0        | 619.0                                                                                | 467.0 | 914.0            | 610.0  | 381.0 | 152.4              | 254.0           |
| 14                | 350 | 355.6            | 533.0          | 222.0 | 279.0 | 358.0     | 165.0        | 711.0                                                                                | 533.0 | 1067.0           | 711.0  | 413.0 | 152.4              | 305.0           |
| 16                | 400 | 406.4            | 610.0          | 254.0 | 305.0 | 406.0     | 178.0        | 813.0                                                                                | 610.0 | 1219.0           | 813.0  | 470.0 | 152.4              | 305.0           |
| 18                | 450 | 457.2            | 686.0          | 286.0 | 343.0 | 457.0     | 203.0        | 914.0                                                                                | 686.0 | 1372.0           | 914.0  | 533.0 | 152.4              | 305.0           |
| 20                | 500 | 508.0            | 762.0          | 318.0 | 381.0 | 508.0     | 229.0        | 1016.0                                                                               | 762.0 | 1524.0           | 1016.0 | 584.0 | 152.4              | 305.0           |
| 22                | 550 | 559.0            | 838.0          | 343.0 | 419.0 | 559.0     | 254.0        | 1118.0                                                                               | 838.0 | 1676.0           | 1118.0 | 614.4 | 152.4              | 305.0           |
| 24                | 600 | 610.0            | 914.0          | 381.0 | 432.0 | 610.0     | 267.0        | 1219.0                                                                               | 914.0 | 1629.0           | 1219.0 | 692.0 | 152.4              | 305.0           |
| 26                | 650 | 660.0            | 991.0          | 405.0 | 495.0 | 660.0     | 267.0        |  |       |                  |        |       |                    |                 |
| 28                | 700 | 711.0            | 1067.0         | 438.0 | 521.0 | 771.0     | 267.0        |                                                                                      |       |                  |        |       |                    |                 |
| 30                | 750 | 762.0            | 1143.0         | 470.0 | 559.0 | 762.0     | 267.0        |                                                                                      |       |                  |        |       |                    |                 |
| 32                | 800 | 813.0            | 1219.0         | 502.0 | 597.0 | 813.0     | 267.0        |                                                                                      |       |                  |        |       |                    |                 |
| 34                | 850 | 864.0            | 1295.0         | 533.0 | 635.0 | 864.0     | 267.0        |                                                                                      |       |                  |        |       |                    |                 |
| 36                | 900 | 914.4            | 1372.0         | 565.0 | 673.0 | 914.0     | 267.0        |                                                                                      |       |                  |        |       |                    |                 |

All Dimension in Millimeters

## BUTT WELD PIPE FITTINGS AS PER ANSI B-16.9



UNEQUAL TEE

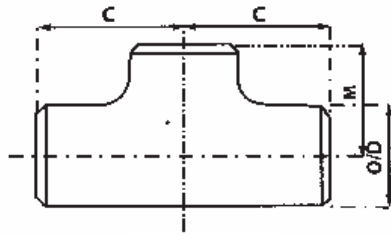


UNEQUAL CROSS

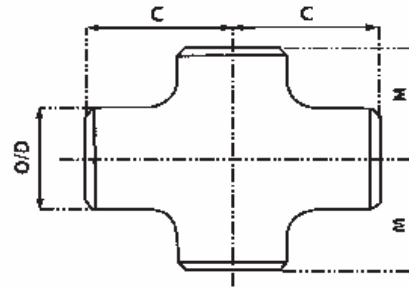
| Nominal Pipe Size (NPS) | Outside Diameter at Bevel (O/D) |        | Center - to - Center |            |
|-------------------------|---------------------------------|--------|----------------------|------------|
|                         | Run                             | Outlet | Run 'C'              | Outlet 'M' |
| 1/2" x 3/8"             | 21.3                            | 17.3   | 25                   | 25         |
| 1/2" x 1/4"             | 21.3                            | 13.7   | 25                   | 25         |
| 3/4" x 1/2"             | 26.7                            | 21.3   | 29                   | 29         |
| 3/4" x 3/8"             | 26.7                            | 17.3   | 29                   | 29         |
| 1" x 3/4"               | 33.4                            | 26.7   | 38                   | 38         |
| 1" x 1/2"               | 33.4                            | 21.3   | 38                   | 38         |
| 1 1/4" x 1"             | 42.2                            | 33.4   | 48                   | 48         |
| 1 1/4" x 3/4"           | 42.2                            | 26.7   | 48                   | 48         |
| 1 1/4" x 1/2"           | 42.2                            | 21.3   | 48                   | 48         |
| 1 1/2" x 1 1/4"         | 48.3                            | 42.2   | 57                   | 57         |
| 1 1/2" x 1"             | 48.3                            | 33.4   | 57                   | 57         |
| 1 1/2" x 3/4"           | 48.3                            | 26.7   | 57                   | 57         |
| 1 1/2" x 1/2"           | 48.3                            | 21.3   | 57                   | 57         |
| 2" x 1 1/2"             | 60.3                            | 48.3   | 64                   | 60         |
| 2" x 1 1/4"             | 60.3                            | 42.2   | 64                   | 57         |
| 2" x 1"                 | 60.3                            | 33.4   | 64                   | 51         |
| 2" x 3/4"               | 60.3                            | 26.7   | 64                   | 44         |
| 2 1/2" x 2"             | 73.0                            | 60.3   | 76                   | 70         |
| 2 1/2" x 1 1/2"         | 73.0                            | 48.3   | 76                   | 67         |
| 2 1/2" x 1 1/4"         | 73.0                            | 42.2   | 76                   | 64         |
| 2 1/2" x 1"             | 73.0                            | 33.4   | 76                   | 57         |
| 3" x 2 1/2"             | 88.9                            | 73.0   | 86                   | 83         |
| 3" x 2"                 | 88.9                            | 60.3   | 86                   | 76         |
| 3" x 1 1/2"             | 88.9                            | 48.3   | 86                   | 73         |
| 3" x 1 1/4"             | 88.9                            | 42.2   | 86                   | 70         |
| 3 1/2" x 3"             | 101.6                           | 88.9   | 95                   | 92         |
| 3 1/2" x 2 1/2"         | 101.6                           | 73.0   | 95                   | 89         |
| 3 1/2" x 2"             | 101.6                           | 60.3   | 95                   | 83         |
| 3 1/2" x 1 1/2"         | 101.6                           | 48.3   | 95                   | 79         |
| 4" x 3 1/2"             | 114.3                           | 101.6  | 105                  | 102        |
| 4" x 3"                 | 114.3                           | 88.9   | 105                  | 98         |
| 4" x 2 1/2"             | 114.3                           | 73.0   | 105                  | 95         |
| 4" x 2"                 | 114.3                           | 60.3   | 105                  | 89         |
| 4" x 1 1/2"             | 114.3                           | 48.3   | 105                  | 86         |
| 5" x 4"                 | 141.3                           | 114.3  | 124                  | 117        |
| 5" x 3 1/2"             | 141.3                           | 101.6  | 124                  | 114        |
| 5" x 3"                 | 141.3                           | 88.9   | 124                  | 111        |
| 5" x 2 1/2"             | 141.3                           | 73.0   | 124                  | 108        |
| 5" x 2"                 | 141.3                           | 60.3   | 124                  | 105        |
| 6" x 5"                 | 168.3                           | 141.3  | 143                  | 137        |
| 6" x 4"                 | 168.3                           | 114.3  | 143                  | 130        |
| 6" x 3 1/2"             | 168.3                           | 101.6  | 143                  | 127        |
| 6" x 3"                 | 168.3                           | 88.9   | 143                  | 124        |
| 6" x 2 1/2"             | 168.3                           | 73.0   | 143                  | 121        |
| 8" x 6"                 | 219.1                           | 168.3  | 178                  | 168        |
| 8" x 5"                 | 218.1                           | 141.3  | 178                  | 162        |
| 8" x 4"                 | 219.1                           | 114.3  | 178                  | 156        |
| 8" x 3 1/2"             | 219.1                           | 101.6  | 178                  | 152        |

| Nominal Pipe Size (NPS) | Outside Diameter at Bevel (O/D) |        | Center - to - Center |            |
|-------------------------|---------------------------------|--------|----------------------|------------|
|                         | Run                             | Outlet | Run 'C'              | Outlet 'M' |
| 10" x 8"                | 273.0                           | 219.1  | 216                  | 203        |
| 10" x 6"                | 273.0                           | 168.3  | 216                  | 194        |
| 10" x 5"                | 273.0                           | 141.3  | 216                  | 191        |
| 10" x 4"                | 273.0                           | 114.3  | 216                  | 184        |
| 12" x 10"               | 323.8                           | 273.0  | 254                  | 241        |
| 12" x 8"                | 323.8                           | 219.1  | 254                  | 229        |
| 12" x 6"                | 323.8                           | 168.3  | 254                  | 219        |
| 12" x 5"                | 323.8                           | 141.3  | 254                  | 216        |
| 14" x 12"               | 355.6                           | 323.8  | 279                  | 270        |
| 14" x 10"               | 355.6                           | 273.0  | 279                  | 257        |
| 14" x 8"                | 355.6                           | 219.1  | 279                  | 248        |
| 14" x 6"                | 355.6                           | 168.3  | 279                  | 238        |
| 16" x 14"               | 406.4                           | 355.6  | 305                  | 305        |
| 16" x 12"               | 406.4                           | 323.8  | 305                  | 295        |
| 16" x 10"               | 406.4                           | 273.0  | 305                  | 283        |
| 16" x 8"                | 406.4                           | 219.1  | 305                  | 273        |
| 16" x 6"                | 406.4                           | 168.3  | 305                  | 264        |
| 18" x 16"               | 457.0                           | 406.4  | 343                  | 330        |
| 18" x 14"               | 457.0                           | 355.6  | 343                  | 330        |
| 18" x 12"               | 457.0                           | 323.8  | 343                  | 321        |
| 18" x 10"               | 457.0                           | 273.0  | 343                  | 308        |
| 18" x 8"                | 457.0                           | 219.1  | 343                  | 298        |
| 20" x 18"               | 508.0                           | 457.0  | 381                  | 368        |
| 20" x 16"               | 508.0                           | 406.4  | 381                  | 356        |
| 20" x 14"               | 508.0                           | 355.6  | 381                  | 356        |
| 20" x 12"               | 508.0                           | 323.8  | 381                  | 346        |
| 20" x 10"               | 508.0                           | 273.0  | 381                  | 333        |
| 20" x 8"                | 508.0                           | 219.1  | 381                  | 324        |
| 22" x 20"               | 559.0                           | 508.0  | 419                  | 406        |
| 22" x 18"               | 559.0                           | 457.0  | 419                  | 394        |
| 22" x 16"               | 559.0                           | 406.4  | 419                  | 381        |
| 22" x 14"               | 559.0                           | 355.6  | 419                  | 381        |
| 22" x 12"               | 559.0                           | 323.8  | 419                  | 371        |
| 22" x 10"               | 559.0                           | 273.0  | 419                  | 359        |
| 24" x 22"               | 610.0                           | 559.0  | 432                  | 432        |
| 24" x 20"               | 610.0                           | 508.0  | 432                  | 432        |
| 24" x 18"               | 610.0                           | 457.0  | 432                  | 419        |
| 24" x 16"               | 610.0                           | 406.4  | 432                  | 406        |
| 24" x 14"               | 610.0                           | 355.6  | 432                  | 406        |
| 24" x 12"               | 610.0                           | 323.8  | 432                  | 397        |
| 24" x 10"               | 610.0                           | 273.0  | 432                  | 384        |
| 26" x 24"               | 660.0                           | 610.0  | 495                  | 483        |
| 26" x 22"               | 660.0                           | 559.0  | 495                  | 470        |
| 26" x 20"               | 660.0                           | 508.0  | 495                  | 457        |
| 26" x 18"               | 660.0                           | 457.0  | 495                  | 444        |
| 26" x 16"               | 660.0                           | 406.4  | 495                  | 432        |
| 26" x 14"               | 660.0                           | 355.6  | 495                  | 432        |
| 26" x 12"               | 660.0                           | 323.8  | 495                  | 422        |
| 28" x 26"               | 711.0                           | 660.0  | 521                  | 521        |
| 28" x 24"               | 711.0                           | 610.0  | 521                  | 508        |
| 28" x 22"               | 711.0                           | 559.0  | 521                  | 495        |

**BUTT WELD PIPE FITTINGS AS PER ANSI B-16.9**



**EQUAL TEE**

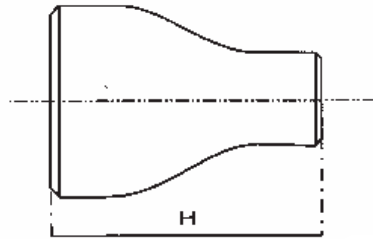


**EQUAL CROSS**

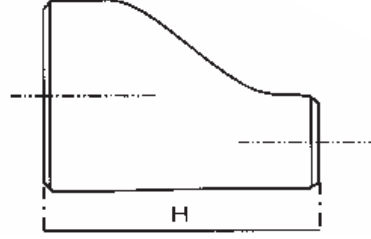
| Nominal Pipe Size (NPS) | Outside Diameter at Bevel (O/D) | Run 'C' | Outlet 'M' |
|-------------------------|---------------------------------|---------|------------|
| 1/2"                    | 21.3                            | 25      | 25         |
| 3/4"                    | 26.7                            | 29      | 29         |
| 1"                      | 33.4                            | 38      | 38         |
| 1.1/4"                  | 42.2                            | 48      | 48         |
| 1.1/2"                  | 48.3                            | 57      | 57         |
| 2"                      | 60.3                            | 64      | 64         |
| 2.1/2"                  | 73.0                            | 76      | 76         |
| 3"                      | 88.9                            | 86      | 86         |
| 3.1/2"                  | 101.6                           | 95      | 95         |
| 4"                      | 114.3                           | 105     | 105        |
| 5"                      | 141.3                           | 124     | 124        |
| 6"                      | 168.3                           | 143     | 143        |
| 8"                      | 219.1                           | 178     | 178        |
| 10"                     | 273.0                           | 216     | 216        |
| 12"                     | 323.8                           | 254     | 254        |
| 14"                     | 355.6                           | 279     | 279        |
| 16"                     | 406.4                           | 305     | 305        |
| 18"                     | 457.0                           | 343     | 343        |
| 20"                     | 508.0                           | 381     | 381        |
| 22"                     | 559.0                           | 419     | 419        |
| 24"                     | 610.0                           | 432     | 432        |
| 26"                     | 660.0                           | 495     | 495        |
| 28"                     | 711.0                           | 521     | 521        |
| 30"                     | 762.0                           | 559     | 559        |
| 32"                     | 813.0                           | 597     | 597        |
| 34"                     | 864.0                           | 635     | 635        |
| 36"                     | 914.0                           | 673     | 673        |

**NOTE : ALL DIMENSIONS ARE IN MM.**

## BUTT WELD PIPE FITTINGS AS PER ANSI B-16.9



CONC. REDUCER



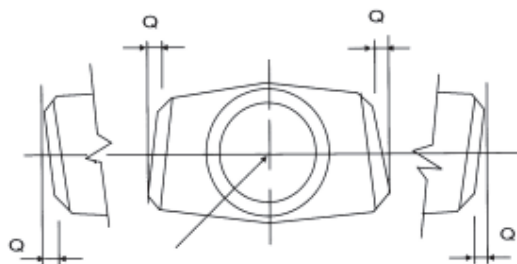
ECC. REDUCER

| Nominal Pipe Size (NPS) | Outside Diameter at Bevel (O/D) |        | End-to-End H |
|-------------------------|---------------------------------|--------|--------------|
|                         | Run                             | Outlet |              |
| 3/4" x 1/2"             | 26.7                            | 21.3   | 38           |
| 3/4" x 3/8"             | 26.7                            | 17.3   | 38           |
| 1" x 3/4"               | 33.4                            | 26.7   | 51           |
| 1" x 1/2"               | 33.4                            | 21.3   | 51           |
| 1.1/4" x 1"             | 42.2                            | 33.4   | 51           |
| 1.1/4" x 3/4"           | 42.2                            | 26.7   | 51           |
| 1.1/4" x 1/2"           | 42.2                            | 21.3   | 51           |
| 1.1/2" x 1.1/4"         | 48.3                            | 42.2   | 64           |
| 1.1/2" x 1"             | 48.3                            | 33.4   | 64           |
| 1.1/2" x 3/4"           | 48.3                            | 26.7   | 64           |
| 1.1/2" x 1/2"           | 46.3                            | 21.3   | 64           |
| 2" x 1.1/2"             | 60.3                            | 48.3   | 76           |
| 2" x 1.1/4"             | 60.3                            | 42.2   | 76           |
| 2" x 1"                 | 60.3                            | 33.4   | 76           |
| 2" x 3/4"               | 60.3                            | 26.7   | 76           |
| 2.1/2" x 2"             | 73.0                            | 60.3   | 89           |
| 2.1/2" x 1.1/2"         | 73.0                            | 48.3   | 89           |
| 2.1/2" x 1.1/4"         | 73.0                            | 42.2   | 89           |
| 2.1/2" x 1"             | 73.0                            | 33.4   | 89           |
| 3" x 2.1/2"             | 88.9                            | 73.0   | 89           |
| 3" x 2"                 | 88.9                            | 60.3   | 89           |
| 3" x 1.1/2"             | 88.9                            | 48.3   | 89           |
| 3" x 1.1/4"             | 88.9                            | 42.2   | 89           |
| 3.1/2" x 3"             | 101.6                           | 88.9   | 102          |
| 3.1/2" x 2.1/2"         | 101.6                           | 73.0   | 102          |
| 3.1/2" x 2"             | 101.6                           | 60.3   | 102          |
| 3.1/2" x 1.1/2"         | 101.6                           | 48.3   | 102          |
| 4" x 3.1/2"             | 114.3                           | 101.6  | 102          |
| 4" x 3"                 | 114.3                           | 88.9   | 102          |
| 4" x 2.1/2"             | 114.3                           | 73.0   | 102          |
| 4" x 2"                 | 114.3                           | 60.3   | 102          |
| 4" x 1.1/2"             | 114.3                           | 48.3   | 102          |
| 5" x 4"                 | 141.3                           | 114.3  | 127          |
| 5" x 3.1/2"             | 141.3                           | 101.6  | 127          |
| 5" x 3"                 | 141.3                           | 88.9   | 127          |
| 5" x 2.1/2"             | 141.3                           | 73.0   | 127          |
| 5" x 2"                 | 141.3                           | 60.3   | 127          |
| 6" x 5"                 | 168.3                           | 141.3  | 140          |
| 6" x 4"                 | 168.3                           | 114.3  | 140          |
| 6" x 3.1/2"             | 168.3                           | 101.6  | 140          |
| 6" x 3"                 | 168.3                           | 88.9   | 140          |
| 6" x 2.1/2"             | 168.3                           | 73.0   | 140          |
| 8" x 6"                 | 219.1                           | 168.3  | 152          |
| 8" x 5"                 | 219.1                           | 141.3  | 152          |
| 8" x 4"                 | 219.1                           | 114.3  | 152          |
| 8" x 3.1/2"             | 219.1                           | 101.6  | 152          |
| 10" x 8"                | 273.0                           | 219.1  | 178          |
| 10" x 6"                | 273.0                           | 168.3  | 178          |
| 10" x 5"                | 273.0                           | 141.3  | 178          |
| 10" x 4"                | 273.0                           | 114.3  | 178          |

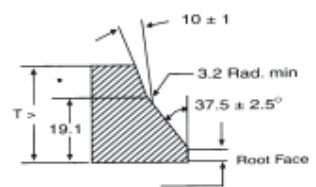
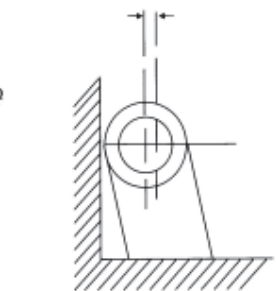
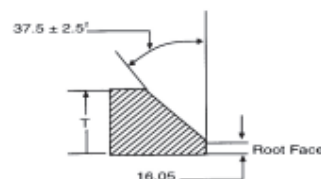
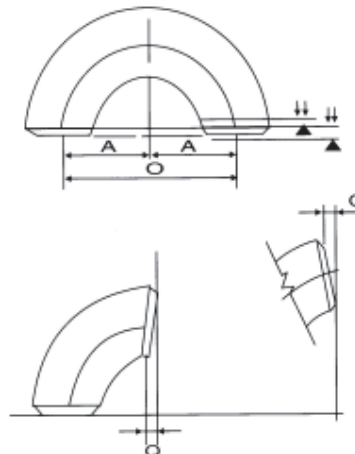
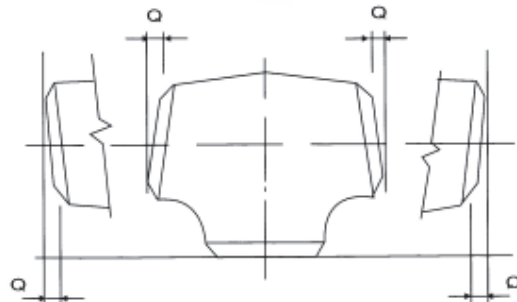
| Nominal Pipe Size (NPS) | Outside Diameter at Bevel (O/D) |        | End-to-End H |
|-------------------------|---------------------------------|--------|--------------|
|                         | Run                             | Outlet |              |
| 12" x 10"               | 323.8                           | 273.0  | 203          |
| 12" x 8"                | 323.8                           | 219.1  | 203          |
| 12" x 6"                | 323.8                           | 168.3  | 203          |
| 12" x 5"                | 323.8                           | 141.3  | 203          |
| 14" x 12"               | 355.6                           | 323.8  | 330          |
| 14" x 10"               | 355.6                           | 273.0  | 330          |
| 14" x 8"                | 355.6                           | 219.1  | 330          |
| 14" x 6"                | 355.6                           | 168.3  | 330          |
| 16" x 14"               | 406.4                           | 355.6  | 356          |
| 16" x 12"               | 406.4                           | 323.8  | 356          |
| 16" x 10"               | 406.4                           | 273.0  | 356          |
| 16" x 8"                | 406.4                           | 219.1  | 356          |
| 18" x 16"               | 457.0                           | 406.4  | 381          |
| 18" x 14"               | 457.0                           | 356.6  | 381          |
| 18" x 12"               | 457.0                           | 323.8  | 381          |
| 18" x 10"               | 457.0                           | 273.0  | 381          |
| 20" x 18"               | 508.0                           | 457.0  | 508          |
| 20" x 16"               | 508.0                           | 406.4  | 508          |
| 20" x 14"               | 508.0                           | 355.6  | 508          |
| 20" x 12"               | 508.0                           | 323.8  | 508          |
| 22" x 20"               | 559.0                           | 508.0  | 508          |
| 22" x 18"               | 559.0                           | 457.0  | 508          |
| 22" x 16"               | 559.0                           | 406.4  | 508          |
| 22" x 14"               | 559.0                           | 355.6  | 508          |
| 24" x 22"               | 610.0                           | 559.0  | 508          |
| 24" x 20"               | 610.0                           | 508.0  | 508          |
| 24" x 18"               | 610.0                           | 457.0  | 508          |
| 24" x 16"               | 610.0                           | 406.4  | 508          |
| 26" x 24"               | 660.0                           | 610.0  | 610          |
| 26" x 22"               | 660.0                           | 559.0  | 610          |
| 26" x 20"               | 660.0                           | 508.0  | 610          |
| 26" x 18"               | 660.0                           | 457.0  | 610          |
| 28" x 26"               | 711.0                           | 660.0  | 610          |
| 28" x 24"               | 711.0                           | 610.0  | 610          |
| 28" x 20"               | 711.0                           | 508.0  | 610          |
| 28" x 18"               | 711.0                           | 457.0  | 610          |
| 30" x 28"               | 762.0                           | 711.0  | 610          |
| 30" x 26"               | 762.0                           | 660.0  | 610          |
| 30" x 24"               | 762.0                           | 610.0  | 610          |
| 30" x 20"               | 762.0                           | 508.0  | 610          |
| 32" x 30"               | 813.0                           | 762.0  | 610          |
| 32" x 28"               | 813.0                           | 711.0  | 610          |
| 32" x 26"               | 813.0                           | 660.0  | 610          |
| 32" x 24"               | 813.0                           | 610.0  | 610          |
| 34" x 32"               | 864.0                           | 813.0  | 610          |
| 34" x 30"               | 864.0                           | 762.0  | 610          |
| 34" x 26"               | 864.0                           | 660.0  | 610          |
| 34" x 24"               | 864.0                           | 610.0  | 610          |

## DIMENSION TOLERANCE ANSI B 16.9 / B16.28 / MSS SP - 43

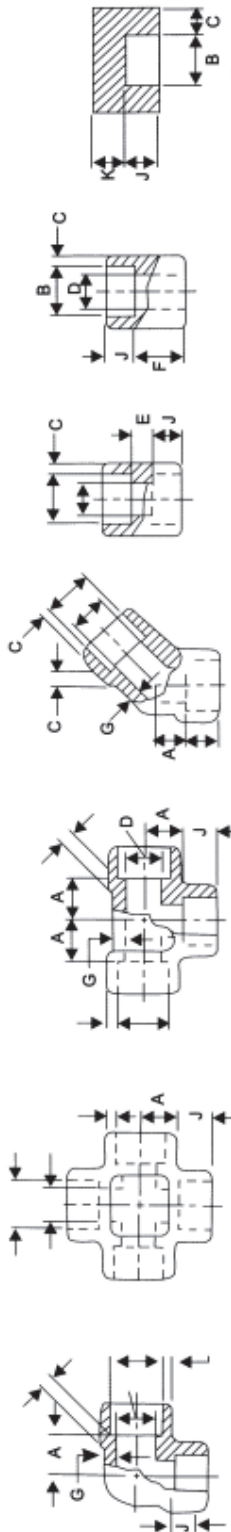
| ALL FITTINGS                 |                           |                |                  |                       |          | 90°/60°/45°<br>/30°<br>ELBOWS<br>& TEES |           | REDUCERS                                 |                           | 180° RETURNS  |          |                         |          |                             |          | CAPS           |          | ANGULARITY TOLERANCE     |                   |                              |           |
|------------------------------|---------------------------|----------------|------------------|-----------------------|----------|-----------------------------------------|-----------|------------------------------------------|---------------------------|---------------|----------|-------------------------|----------|-----------------------------|----------|----------------|----------|--------------------------|-------------------|------------------------------|-----------|
| Nominal Pipe Size<br>Inch/mm | Outside Diameter at Level |                | Inside Dia meter | Wall Thickness at End |          | Centre to End                           |           | Overall Length Dimensions                |                           | Centre to End |          | Back to Face Dimensions |          | Alignment of End Dimensions |          | Overall length |          | Nominal Pipe Size        | Off Angle Inch/mm |                              | Off Plane |
| D                            |                           |                |                  | T                     |          | A,B,C,M                                 |           | H                                        |                           | O             |          | K                       |          | U                           |          | E              |          |                          | Q                 |                              | P         |
|                              | (1)<br>B16.9              | MSS SP43       | (2)<br>B16.9     | B16.9                 | MSS SP43 | B16.9                                   | MSS SP43  | B16.9                                    | MSS SP43                  | B16.9         | MSS SP43 | B16.9                   | MSS SP43 | B16.9                       | MSS SP43 | B16.9          | MSS SP43 |                          | B16.9             | MSS SP43                     | B16.9     |
| 1/2" -2 1/2"<br>15 - 65      | ±1.6<br>- 0.8             |                | ±0.8             |                       |          | FROM 1/2" TO 18"<br>15 TO 600           | FROM 3/4" | FROM 1/2"-24"<br>15 - 600                | FROM 1/2"-28"<br>15 - 200 |               |          |                         |          |                             |          | ±3             | ±3.17    | 1/2" - 4"<br>15 - 100    | ±1                |                              | ±2        |
| 3" - 3 1/2"<br>80 - 90       |                           |                |                  |                       |          |                                         |           |                                          |                           |               |          |                         |          |                             |          |                |          | 5" - 8"<br>125 - 200     | ±2                |                              | ±4        |
| 4"<br>100                    | ±1.6                      |                |                  |                       |          |                                         |           |                                          |                           |               |          |                         |          |                             |          |                |          | 10" - 12"<br>250 - 300   | ±3                |                              | ±5        |
| 5" - 6"<br>125 - 150         |                           |                |                  |                       |          |                                         |           |                                          |                           |               |          |                         |          |                             |          |                |          | 14" - 16"<br>350 - 400   | ±3                |                              | ±7        |
| 8"<br>200                    | ±2.4<br>-1.6              |                |                  |                       |          |                                         |           |                                          |                           |               |          |                         |          |                             |          |                |          | 18" - 24"<br>450 - 600   | ±3                |                              | ±10       |
| 10" - 18"<br>250 - 450       | +4<br>-3.2                | +2.38<br>-0.80 | ±3.2             |                       |          |                                         | ±2.40     |                                          |                           |               |          |                         |          |                             |          |                |          | 26" - 30"<br>650 - 750   | ±4                |                              | ±10       |
| 20" - 24"<br>500 - 600       | +6.4<br>-4.8              | 3.17<br>0.79   |                  |                       |          |                                         |           |                                          |                           | ±10"          | ±10"     |                         |          | ±2.0"                       | ±1.60    |                |          | 32" - 42"<br>800 - 1050  | ±5                |                              | ±13       |
| 26"- 30"<br>650 - 750        | +6.4<br>-4.8              | ±4.8           | ±4.8             |                       |          | ±3                                      |           | FROM 26"- 48"<br>650NB - 701200 NB<br>±5 |                           |               |          |                         |          |                             |          |                | ±10      | 44" - 48"<br>1100 - 1200 | ±5                | 38" - 48"<br>950-1200<br>3.2 | ±20       |
| 32" -48"<br>800 - 1200       | +6.4<br>-4.8              |                |                  |                       |          | ±5                                      |           |                                          |                           |               |          |                         |          |                             |          |                |          | 42" - 48"<br>1050 - 1200 | ±5                |                              | ±20       |



This End Flush Against Square



## DIMENSIONS OF SOCKET-WELD FITTINGS ANSI B 16.11



| 90°Elbow  | Cross             |                        |                                       | Tee                                 |      | 45° Elbow |             | Coupling               |                               | Half Coupling |       | Cap            |                  |            |     |                          |                   |      |      |      |      |      |      |      |     |     |     |      |      |      |
|-----------|-------------------|------------------------|---------------------------------------|-------------------------------------|------|-----------|-------------|------------------------|-------------------------------|---------------|-------|----------------|------------------|------------|-----|--------------------------|-------------------|------|------|------|------|------|------|------|-----|-----|-----|------|------|------|
|           | Nominal Pipe Size | Socket Bore Diameter B | Bore Diameter of Fitting D [Note (1)] | Socket Wall Thickness, C [Note (2)] |      |           | Body wall G | Min. Depth of Socket J | Center to Bottom of Socket, A |               |       | Laying Lengths |                  | Tolerances |     | End Wall Thickness K Min |                   |      |      |      |      |      |      |      |     |     |     |      |      |      |
|           |                   |                        |                                       | Class Designation                   |      |           |             |                        | Class Designation             |               |       | Couplings E    | Half Couplings F | A          | E   | F                        | Class Designation |      |      |      |      |      |      |      |     |     |     |      |      |      |
|           |                   |                        |                                       | 3000                                | 6000 | 9000      |             |                        | 3000                          | 6000          | 9000  |                |                  |            |     |                          | 3000              | 6000 | 9000 |      |      |      |      |      |     |     |     |      |      |      |
| Avg. Min. |                   |                        |                                       |                                     |      |           |             |                        |                               |               |       |                |                  |            |     |                          |                   |      |      |      |      |      |      |      |     |     |     |      |      |      |
| 6         | 1/8               | 11.2                   | 7.6                                   | 4.8                                 | ...  | ...       | 3.18        | 3.18                   | 3.96                          | 3.43          | ...   | ...            | 2.41             | 3.15       | ... | 9.5                      | 11.0              | 11.0 | ...  | 8.0  | 8.0  | ...  | 6.5  | 16.0 | 1.0 | 1.5 | 1.0 | 4.8  | 6.4  | ...  |
|           |                   | 10.8                   | 6.1                                   | 3.2                                 | ...  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 8         | 1/4               | 14.6                   | 10.0                                  | 7.1                                 | ...  | ...       | 3.78        | 3.30                   | 4.60                          | 4.01          | ...   | ...            | 3.02             | 3.68       | ... | 9.5                      | 11.0              | 13.5 | ...  | 8.0  | 8.0  | ...  | 6.5  | 16.0 | 1.0 | 1.5 | 1.0 | 4.8  | 6.4  | ...  |
|           |                   | 14.2                   | 8.5                                   | 5.6                                 | ...  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 10        | 3/8               | 18.0                   | 13.3                                  | 9.9                                 | ...  | ...       | 4.01        | 3.50                   | 5.03                          | 4.37          | ...   | ...            | 3.20             | 4.01       | ... | 9.5                      | 13.5              | 15.5 | ...  | 8.0  | 11.0 | ...  | 6.5  | 17.5 | 1.5 | 3.0 | 1.5 | 4.8  | 6.4  | ...  |
|           |                   | 17.6                   | 11.8                                  | 8.4                                 | ...  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 15        | 1/2               | 22.2                   | 16.6                                  | 12.5                                | 7.2  | 4.67      | 4.09        | 5.97                   | 5.18                          | 9.35          | 8.18  | 3.73           | 4.78             | 7.47       | ... | 9.5                      | 15.5              | 19.5 | 25.5 | 11.0 | 12.5 | 15.5 | 9.5  | 22.5 | 1.5 | 3.0 | 1.5 | 6.4  | 7.9  | 11.2 |
|           |                   | 21.8                   | 15.0                                  | 11.0                                | 5.6  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 20        | 3/4               | 27.6                   | 21.7                                  | 16.3                                | 11.8 | 4.90      | 4.27        | 6.96                   | 6.04                          | 9.78          | 8.56  | 3.91           | 5.56             | 7.82       | ... | 12.5                     | 19.0              | 22.5 | 28.5 | 13.0 | 14.0 | 19.0 | 9.5  | 24.0 | 1.5 | 3.0 | 1.5 | 6.4  | 7.9  | 12.7 |
|           |                   | 27.2                   | 20.2                                  | 14.8                                | 10.3 | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 25        | 1                 | 34.3                   | 27.4                                  | 21.5                                | 16.0 | 5.69      | 4.98        | 7.92                   | 6.93                          | 11.38         | 9.96  | 4.55           | 6.35             | 9.09       | ... | 12.5                     | 22.5              | 27.0 | 32.0 | 14.0 | 17.5 | 20.5 | 12.5 | 28.5 | 2.0 | 4.0 | 2.0 | 9.6  | 11.2 | 14.2 |
|           |                   | 33.9                   | 25.9                                  | 19.9                                | 14.4 | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 32        | 1 1/4             | 43.1                   | 35.8                                  | 30.2                                | 23.5 | 6.07      | 5.28        | 7.92                   | 6.93                          | 12.14         | 10.62 | 4.85           | 6.35             | 9.70       | ... | 12.5                     | 27.0              | 32.0 | 35.0 | 17.5 | 20.5 | 22.5 | 12.5 | 30.0 | 2.0 | 4.0 | 2.0 | 9.6  | 11.2 | 14.2 |
|           |                   | 42.7                   | 34.3                                  | 28.7                                | 22.0 | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 40        | 1 1/2             | 49.2                   | 41.6                                  | 34.7                                | 28.7 | 6.35      | 5.54        | 8.92                   | 7.80                          | 12.70         | 11.12 | 5.08           | 7.14             | 10.15      | ... | 12.5                     | 32.0              | 38.0 | 38.0 | 20.5 | 25.5 | 25.5 | 12.5 | 32.0 | 2.0 | 4.0 | 2.0 | 11.2 | 12.7 | 15.7 |
|           |                   | 48.8                   | 40.1                                  | 33.2                                | 27.2 | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 50        | 2                 | 61.7                   | 53.3                                  | 43.6                                | 38.9 | 6.93      | 6.04        | 10.92                  | 9.50                          | 13.84         | 12.12 | 5.54           | 8.74             | 11.07      | ... | 16.0                     | 38.0              | 41.0 | 54.0 | 25.5 | 28.5 | 28.5 | 19.0 | 41.0 | 2.0 | 4.0 | 2.0 | 12.7 | 15.7 | 19.0 |
|           |                   | 61.2                   | 51.7                                  | 42.1                                | 37.4 | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 65        | 2 1/2             | 74.4                   | 64.2                                  | ...                                 | ...  | ...       | 8.76        | 7.67                   | ...                           | ...           | ...   | ...            | 7.01             | ...        | ... | 16.0                     | 41.0              | ...  | ...  | 28.5 | ...  | ...  | 19.0 | 43.0 | 2.5 | 5.0 | 2.5 | 15.7 | 19.0 | ...  |
|           |                   | 73.9                   | 61.2                                  | ...                                 | ...  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 80        | 3                 | 90.3                   | 79.4                                  | ...                                 | ...  | ...       | 9.52        | 8.30                   | ...                           | ...           | ...   | ...            | 7.62             | ...        | ... | 16.0                     | 57.0              | ...  | ...  | 32.0 | ...  | ...  | 19.0 | 44.5 | 2.5 | 5.0 | 2.5 | 19.0 | 22.4 | ...  |
|           |                   | 89.8                   | 76.4                                  | ...                                 | ...  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |
| 100       | 4                 | 115.7                  | 103.8                                 | ...                                 | ...  | ...       | 10.69       | 9.35                   | ...                           | ...           | ...   | ...            | 8.56             | ...        | ... | 19.0                     | 66.5              | ...  | ...  | 41.0 | ...  | ...  | 19.0 | 48.0 | 2.5 | 5.0 | 2.5 | 22.4 | 28.4 | ...  |
|           |                   | 115.2                  | 100.7                                 | ...                                 | ...  | ...       | ...         | ...                    | ...                           | ...           | ...   | ...            | ...              | ...        | ... | ...                      | ...               | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ... | ... | ... | ...  | ...  | ...  |

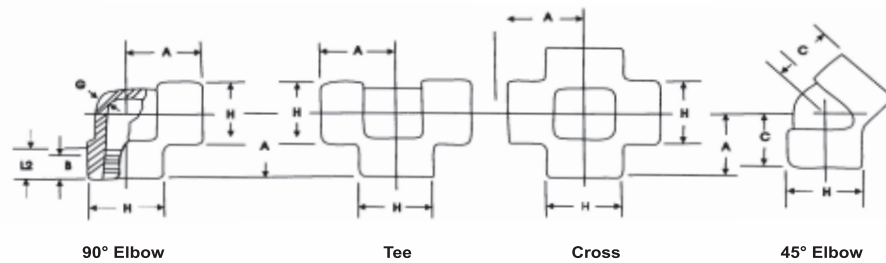
Avg. Min.

(1) All Dimensions Are In Millimeters

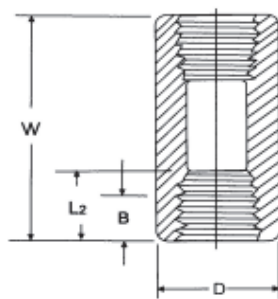
(2) Upper and lower values for each size are the respective maximum and minimum dimensions.

(3) Average of socket wall thickness around periphery shall be no less than listed values. The minimum values are permitted in localized areas.

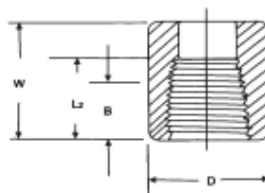
## DIMENSIONS OF THREADED FITTINGS ANSI B 16.11



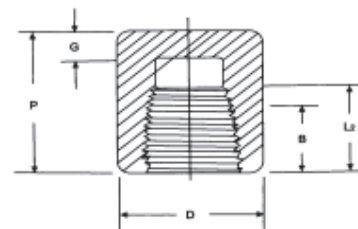
| Nominal Pipe Size |       | Center to End Elbows Tees, Crosses A |      |      | Center to End 45 Elbows C |      |      | Outside Diameter of Bend H |      |      | Minimum Wall Thickness G |       |       | Min. Length of Thread Note - 1 |      |
|-------------------|-------|--------------------------------------|------|------|---------------------------|------|------|----------------------------|------|------|--------------------------|-------|-------|--------------------------------|------|
| NB                | INCH  | 2000                                 | 3000 | 6000 | 2000                      | 3000 | 6000 | 2000                       | 3000 | 6000 | 2000                     | 3000  | 6000  | B                              | L2   |
| 6                 | 1/8   | 21                                   | 21   | 25   | 17                        | 17   | 19   | 22                         | 22   | 25   | 3.18                     | 3.18  | 6.35  | 6.4                            | 6.7  |
| 8                 | 1/4   | 21                                   | 25   | 28   | 17                        | 19   | 22   | 22                         | 25   | 33   | 3.18                     | 3.30  | 6.60  | 8.1                            | 10.2 |
| 10                | 3/8   | 25                                   | 28   | 33   | 19                        | 22   | 25   | 25                         | 33   | 38   | 3.18                     | 3.51  | 6.98  | 9.1                            | 10.4 |
| 15                | 1/2   | 28                                   | 33   | 38   | 22                        | 25   | 28   | 33                         | 38   | 46   | 3.18                     | 4.09  | 8.15  | 10.9                           | 13.6 |
| 20                | 3/4   | 33                                   | 38   | 44   | 25                        | 28   | 33   | 38                         | 46   | 56   | 3.18                     | 4.32  | 8.53  | 12.7                           | 13.9 |
| 25                | 1     | 38                                   | 44   | 51   | 28                        | 33   | 35   | 46                         | 56   | 62   | 3.68                     | 4.98  | 9.93  | 14.7                           | 17.3 |
| 32                | 1 1/4 | 44                                   | 51   | 60   | 33                        | 35   | 43   | 56                         | 62   | 75   | 3.89                     | 5.28  | 10.59 | 17.0                           | 18.0 |
| 40                | 1 1/2 | 51                                   | 60   | 64   | 35                        | 43   | 44   | 62                         | 75   | 84   | 4.01                     | 5.56  | 11.07 | 17.8                           | 18.4 |
| 50                | 2     | 60                                   | 64   | 83   | 43                        | 44   | 52   | 75                         | 84   | 102  | 4.27                     | 7.14  | 12.09 | 19.0                           | 19.2 |
| 65                | 2 1/2 | 76                                   | 83   | 95   | 52                        | 52   | 64   | 92                         | 102  | 121  | 5.61                     | 7.65  | 15.29 | 23.6                           | 28.9 |
| 80                | 3     | 86                                   | 95   | 106  | 64                        | 64   | 79   | 109                        | 121  | 146  | 5.99                     | 8.84  | 16.64 | 25.9                           | 30.5 |
| 100               | 4     | 106                                  | 114  | 114  | 79                        | 79   | 79   | 146                        | 152  | 152  | 6.55                     | 11.18 | 18.67 | 27.7                           | 33.0 |



Coupling



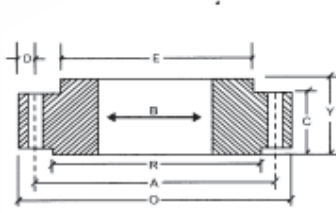
Half-Coupling



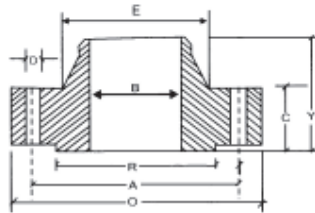
Cap

| Nominal Pipe Size |       | End to End Couplings W | End to End Caps P |      | Outside Diameter D |      | End Wall Thickness min. G |      | Min. Length of Thread Note - 1 |      |
|-------------------|-------|------------------------|-------------------|------|--------------------|------|---------------------------|------|--------------------------------|------|
| NB                | INCH  | 3000 and 6000          | 3000              | 6000 | 3000               | 6000 | 3000                      | 6000 | B                              | L2   |
| 6                 | 1/8   | 32                     | 19                | -    | 16                 | 22   | 4.8                       | -    | 6.4                            | 6.7  |
| 8                 | 1/4   | 35                     | 25                | 27   | 19                 | 25   | 4.8                       | 6.4  | 8.1                            | 10.2 |
| 10                | 3/8   | 38                     | 25                | 27   | 22                 | 32   | 4.8                       | 6.4  | 9.1                            | 10.4 |
| 15                | 1/2   | 48                     | 32                | 33   | 28                 | 38   | 6.4                       | 7.9  | 10.9                           | 13.6 |
| 20                | 3/4   | 51                     | 37                | 38   | 35                 | 44   | 6.4                       | 7.9  | 12.7                           | 13.9 |
| 25                | 1     | 60                     | 41                | 43   | 44                 | 57   | 9.7                       | 11.2 | 14.7                           | 17.3 |
| 32                | 1 1/4 | 67                     | 44                | 46   | 57                 | 64   | 9.7                       | 11.2 | 17.0                           | 18.0 |
| 40                | 1 1/2 | 79                     | 44                | 48   | 64                 | 76   | 11.2                      | 12.7 | 17.8                           | 18.4 |
| 50                | 2     | 86                     | 48                | 51   | 76                 | 92   | 12.7                      | 15.7 | 19.0                           | 19.2 |
| 65                | 2 1/2 | 92                     | 60                | 64   | 92                 | 108  | 15.7                      | 19.0 | 23.6                           | 28.9 |
| 80                | 3     | 108                    | 65                | 68   | 108                | 127  | 19.0                      | 22.4 | 25.9                           | 30.5 |
| 100               | 4     | 121                    | 68                | 75   | 140                | 159  | 22.4                      | 28.4 | 27.7                           | 33.0 |

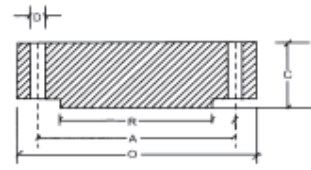
## DIMENSIONS OF FORGED FLANGES ANSI B 16.5



SLIP-ON FLANGE



WELDING NECK FLANGE



BLIND FLANGE

### ASA 150 CLASS

| Nominal<br>Pipe<br>Size<br>(MM) (INCH.) | Flange<br>Dia<br>O | Dia of<br>Bolt<br>Circle<br>A | Dia of<br>Bolt<br>Holes<br>D | No. of<br>Holes | Thk of<br>Flange<br>C | Dia of<br>Hub<br>E | Length through Hub |          |          | Dia of Bore    |          | Dia of<br>R/F<br>R | Depth of<br>Socket<br>F | Pipe<br>Dia<br>X |
|-----------------------------------------|--------------------|-------------------------------|------------------------------|-----------------|-----------------------|--------------------|--------------------|----------|----------|----------------|----------|--------------------|-------------------------|------------------|
|                                         |                    |                               |                              |                 |                       |                    | S/O & S/W<br>Y     | W/N<br>Y | L/J<br>Y | S/O & S/W<br>B | L/J<br>B |                    |                         |                  |
| 15                                      | 1/2                | 90.0                          | 60.3                         | 15.9            | 4                     | 11.1               | 15.9               | 47.6     | 15.9     | 22.3           | 22.9     | 34.9               | 9.5                     | 21.33            |
| 20                                      | 3/4                | 100.0                         | 69.8                         | 15.9            | 4                     | 12.7               | 15.9               | 52.4     | 15.9     | 27.7           | 28.2     | 42.9               | 11.1                    | 26.67            |
| 25                                      | 1                  | 110.0                         | 79.4                         | 15.9            | 4                     | 14.3               | 17.5               | 55.6     | 17.5     | 34.5           | 35.0     | 50.8               | 12.7                    | 33.40            |
| 32                                      | 1 1/4              | 115.0                         | 88.9                         | 15.9            | 4                     | 15.9               | 20.6               | 57.1     | 20.6     | 43.2           | 43.7     | 63.5               | 14.3                    | 42.16            |
| 40                                      | 1 1/2              | 125.0                         | 98.4                         | 15.9            | 4                     | 17.5               | 22.2               | 61.9     | 22.2     | 49.5           | 50.0     | 73.0               | 15.9                    | 48.26            |
| 50                                      | 2                  | 150.0                         | 120.6                        | 19.0            | 4                     | 19.0               | 25.4               | 63.5     | 25.4     | 62.0           | 62.5     | 92.1               | 17.5                    | 60.31            |
| 65                                      | 2 1/2              | 180.0                         | 139.7                        | 19.0            | 4                     | 22.2               | 28.6               | 69.8     | 28.6     | 74.7           | 75.4     | 104.8              | 19.0                    | 73.02            |
| 80                                      | 3                  | 190.0                         | 152.4                        | 19.0            | 4                     | 23.8               | 30.2               | 69.8     | 30.2     | 90.7           | 91.4     | 127.0              | 20.6                    | 88.90            |
| 100                                     | 4                  | 230.0                         | 190.5                        | 19.0            | 8                     | 23.8               | 33.3               | 76.2     | 33.3     | 116.1          | 116.8    | 157.2              | 23.8                    | 114.30           |
| 125                                     | 5                  | 255.0                         | 215.9                        | 22.2            | 8                     | 23.8               | 36.5               | 88.9     | 36.5     | 143.8          | 144.5    | 185.7              | 23.8                    | 141.30           |
| 150                                     | 6                  | 280.0                         | 241.3                        | 22.2            | 8                     | 25.4               | 39.7               | 88.9     | 39.7     | 170.7          | 171.4    | 215.9              | 27.0                    | 168.27           |
| 200                                     | 8                  | 345.0                         | 298.4                        | 22.2            | 8                     | 28.6               | 44.4               | 101.6    | 44.4     | 221.5          | 222.2    | 269.9              | 31.7                    | 219.07           |
| 250                                     | 10                 | 405.0                         | 361.9                        | 25.4            | 12                    | 30.2               | 49.2               | 101.6    | 49.2     | 276.3          | 277.4    | 323.8              | 33.3                    | 273.05           |
| 300                                     | 12                 | 485.0                         | 431.8                        | 25.4            | 12                    | 31.8               | 55.6               | 114.3    | 55.6     | 327.1          | 328.2    | 381.0              | 39.7                    | 323.85           |
| 350                                     | 14                 | 535.0                         | 476.2                        | 28.6            | 12                    | 34.9               | 57.1               | 127.0    | 79.4     | 359.1          | 360.2    | 412.7              | 41.3                    | 355.60           |
| 400                                     | 16                 | 595.0                         | 539.7                        | 28.6            | 16                    | 36.5               | 63.5               | 127.0    | 87.3     | 410.5          | 411.2    | 469.9              | 44.4                    | 406.40           |
| 450                                     | 18                 | 635.0                         | 577.8                        | 31.7            | 16                    | 39.7               | 68.3               | 139.7    | 96.8     | 461.8          | 462.3    | 533.4              | 49.2                    | 457.20           |
| 500                                     | 20                 | 700.0                         | 635.0                        | 31.7            | 20                    | 42.9               | 73.0               | 144.5    | 103.2    | 513.1          | 514.3    | 584.2              | 54.0                    | 508.00           |
| 600                                     | 24                 | 815.0                         | 749.3                        | 34.9            | 20                    | 47.6               | 82.5               | 152.4    | 111.1    | 615.9          | 615.9    | 692.1              | 63.5                    | 609.60           |

All Dimensions are in Millimeters • Flanges except Lap Joint will be furnished with (1.6mm) Raised Face, which is included in Thickness(C) and Length through Hub(Y).

### ASA 300 CLASS

| Nominal<br>Pipe<br>Size<br>(MM) (INCH.) | Flange<br>Dia<br>O | Dia of<br>Bolt<br>Circle<br>A | Dia of<br>Bolt<br>Holes<br>D | No. of<br>Holes | Thk of<br>Flange<br>C | Dia of<br>Hub<br>E | Length through Hub |          |          | Dia of Bore    |          | Dia of<br>R/F<br>R | Depth of<br>Socket<br>F | Pipe<br>Dia<br>X |
|-----------------------------------------|--------------------|-------------------------------|------------------------------|-----------------|-----------------------|--------------------|--------------------|----------|----------|----------------|----------|--------------------|-------------------------|------------------|
|                                         |                    |                               |                              |                 |                       |                    | S/O & S/W<br>Y     | W/N<br>Y | L/J<br>Y | S/O & S/W<br>B | L/J<br>B |                    |                         |                  |
| 15                                      | 1/2                | 95.0                          | 66.7                         | 15.9            | 4                     | 14.3               | 22.2               | 52.4     | 22.2     | 22.3           | 22.9     | 34.9               | 9.5                     | 21.33            |
| 20                                      | 3/4                | 115.0                         | 82.5                         | 19.0            | 4                     | 15.9               | 25.4               | 57.1     | 25.4     | 27.7           | 28.2     | 42.9               | 11.1                    | 26.67            |
| 25                                      | 1                  | 125.0                         | 88.9                         | 19.0            | 4                     | 17.5               | 27.0               | 61.9     | 27.0     | 34.5           | 35.0     | 50.8               | 12.7                    | 33.40            |
| 32                                      | 1 1/4              | 135.0                         | 98.4                         | 19.0            | 4                     | 19.0               | 27.0               | 65.1     | 27.0     | 43.2           | 43.7     | 63.5               | 14.3                    | 42.16            |
| 40                                      | 1 1/2              | 155.0                         | 114.3                        | 22.2            | 4                     | 20.6               | 30.2               | 68.3     | 30.2     | 49.5           | 50.0     | 73.0               | 15.9                    | 48.26            |
| 50                                      | 2                  | 165.0                         | 127.0                        | 19.0            | 8                     | 22.2               | 33.3               | 69.8     | 33.3     | 62.0           | 62.5     | 92.1               | 17.5                    | 60.31            |
| 65                                      | 2 1/2              | 190.0                         | 149.2                        | 22.2            | 8                     | 25.4               | 38.1               | 76.2     | 38.1     | 74.7           | 75.4     | 104.8              | 19.0                    | 73.02            |
| 80                                      | 3                  | 210.0                         | 168.3                        | 22.2            | 8                     | 28.6               | 42.9               | 79.4     | 42.9     | 90.7           | 91.4     | 127.0              | 20.6                    | 88.90            |
| 100                                     | 4                  | 255.0                         | 200.0                        | 22.2            | 8                     | 31.8               | 47.6               | 85.7     | 47.6     | 116.1          | 116.8    | 157.2              | 23.8                    | 114.30           |
| 125                                     | 5                  | 280.0                         | 234.9                        | 22.2            | 8                     | 34.9               | 50.8               | 98.4     | 50.8     | 143.8          | 144.5    | 185.7              | -                       | 141.30           |
| 150                                     | 6                  | 320.0                         | 269.9                        | 22.2            | 12                    | 36.5               | 52.4               | 98.4     | 52.4     | 170.7          | 171.4    | 215.9              | -                       | 168.27           |
| 200                                     | 8                  | 380.0                         | 330.2                        | 25.4            | 12                    | 41.3               | 61.9               | 111.1    | 61.9     | 221.5          | 222.2    | 269.9              | -                       | 219.07           |
| 250                                     | 10                 | 445.0                         | 387.3                        | 28.6            | 16                    | 47.6               | 66.7               | 117.5    | 95.2     | 276.3          | 277.4    | 323.8              | -                       | 273.05           |
| 300                                     | 12                 | 520.0                         | 450.8                        | 31.7            | 16                    | 50.8               | 73.0               | 130.2    | 101.6    | 327.1          | 328.2    | 381.0              | -                       | 323.85           |
| 350                                     | 14                 | 585.0                         | 514.3                        | 31.7            | 20                    | 54.0               | 76.2               | 142.9    | 111.1    | 359.1          | 360.2    | 412.7              | -                       | 355.60           |
| 400                                     | 16                 | 650.0                         | 571.5                        | 34.9            | 20                    | 57.2               | 82.5               | 146.0    | 120.6    | 410.5          | 411.2    | 469.9              | -                       | 406.40           |
| 450                                     | 18                 | 710.0                         | 628.5                        | 34.9            | 24                    | 60.3               | 88.9               | 158.7    | 130.2    | 461.8          | 462.3    | 533.4              | -                       | 457.20           |
| 500                                     | 20                 | 775.0                         | 685.8                        | 34.9            | 24                    | 63.5               | 95.2               | 161.9    | 139.7    | 513.1          | 514.3    | 584.2              | -                       | 508.00           |
| 600                                     | 24                 | 915.0                         | 812.8                        | 41.3            | 24                    | 69.8               | 106.4              | 168.3    | 152.4    | 615.9          | 615.9    | 692.1              | -                       | 609.60           |

All Dimensions are in Millimeters • Flanges except Lap Joint will be furnished with (1.6mm) Raised Face, which is included in Thickness(C) and Length through Hub(Y).